

‘Affectibility’ and Design Workshops: Taking actions towards more sensible design

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ABSTRACT

The understanding of subjective themes like affect and emotions constitute a challenge in interaction design, especially in the field of digital technology to support learning environments. Although literature concerning affective design presents varied results, there is still a lack of works with techniques and methods to support designers in their practice. In this work we present a practical approach based on Framework and Principles for the Design for Affectibility – the design concerned with affective responses from users in their interaction with technology. The design of an application is illustrated within users’ environment and with their active participation. Preliminary evaluation of such the application suggests that the principles can be a favorable instrument to bring affect considerations into the design of technology for learning environments.

Author Keywords

Affectibility; Design for, with and by children; affect and emotions in design; learning technology; OLPC.

ACM Classification Keywords

H.1.2 [User/Machine Systems]: Human factors; K.3.2 [Computers and Education]: Computer uses in education.

General Terms

Design, Human Factors.

INTRODUCTION

The use of Information and Communication Technology may have a positive effect on education [38]. While the technology alone has little power to improve education, its meaningful use aligned with the learners’ context and interests might be of greater value. Some applications evoke rather positive affective responses during users’ interaction with it, while others are not so successful. The understanding of what might contribute to such affective results could empower designers and educators in the

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creation of more appealing and engaging technology.

Researchers from different areas have been interested in the roles that affect plays. In the area of biological sciences, studies suggested a relationship between negative affective attitudes and weaker immune responses [34]. In contrast, experiments from the field of psychology have suggested that positive affect might facilitate creative problem solving [22]. Authorities in the area of psychology and education have emphasized the importance of affect in children’s development [24]. For example, for Piaget, affectivity is an energetic source, like the fuel that makes the motor of intelligence run without modifying the motor’s structure [6,24].

The HCI community has also shown a crescent interest in the subject, aiming at understanding the affective responses of users and its implications to interaction design, as we can see in works coming from different contexts and different target audiences [e.g.,13,39]. Despite this interest, there is still a lack of studies regarding practical approaches to support the design aiming at more positive affective responses, especially in learning contexts. While some specific approaches can be found (e.g., in the creation of animations [13] or systems for geo-simulation [11]), research on general design principles is still needed [17].

In this paper we present an approach to explicitly include, in the design process, the concern with affective aspects of interaction. This paper illustrates the approach with design workshops that took place at a public school in Brazil. Children, actively creating interface elements and generating ideas in these workshops, were our partners in the design of an educational application. Low-fidelity prototypes of this application were explored in a preliminary evaluation performed by children themselves.

This paper is organized as follows: The next Section presents related works, followed by the Framework and Principles regarding the Design for Affectibility. Then we describe the Design Workshops, as part of practical application of our Framework and we present the results of the Design Workshops, discussing them. The last section concludes pointing out further work.

DESIGN WITH CHILDREN AND AFFECT

A rich and interesting literature can be found on design with/by children. Most cited works include those from Druin [7] and Antle [1]. Other authors also report on their

experiences with design and participatory practices with children. There are works that emphasize certain methods as, for example, narratives [8,13], storytelling [29], and other activities concerned with children's context and culture [25]. Other authors investigate the specificity of designing with children with dyslexia [32], autism [10,28] and motor impairment [20].

As mentioned before, affect plays an important role in students' cognitive development – as it has been argued since long by authors like Vygotsky, Piaget and Wallon [24]. However, models and frameworks that inform how to deal with affect in the practice of design are still needed.

Affect can be understood by different authors in different ways. Almost 100 different definitions were gathered for the words affect and emotion [23]. Discussions on meanings for the words 'affect' and 'emotion' are not in the scope of our work; we adopt one of the definitions with the purpose of making it clear what we mean. When we refer to 'affect' we consider Norman's definition in which affect is *"a superordinate concept that subsumes particular valenced conditions such as emotions, moods, feelings, and preferences."* It is a *"general construct that encompasses a wide range of psychological conditions relating to value"* [31].

Moreover, our work is aligned to the concept of "interactional" approach to affect [5]. In this approach, affect is seen as a product of society, where people's cultures and values are important to construct the specific meaning emotions have for certain groups of users. On an opposite direction, the authors call "informational" those approaches that see affect and emotion as objective units that are internally (individually) constructed and subsequently delivered. Propositions that are aligned with such "informational" approach are those systems that automatically recognize and interpret emotions. Although such approaches are not the focus of our work, we believe that, to some extent, the "informational" approaches can also contribute to the Design for Affectability. Affectability [15] is related to the characteristics of a digital artifact that help in triggering emotional and affective states during users' interactions with it. It refers to the aspects of systems that make them more likely to evoke certain affective responses, including emotional, hedonic, aesthetic, etc. The term Affectability goes further than the term Affectibility (with an 'a' instead of an 'i') used by Hoonhout [19], which considers mainly the fun facet (the hedonic qualities) of the matter.

Research can be found that concerns the development of applications that are specific for dealing with affect within learning environments. For example, Girard et al., [13] propose a set of recommendations for the design of affective expressions in digital characters. The authors design and evaluate, together with students, animations to be included in a learning environment. We propose principles that could be considered to be on a rather higher

level of abstraction, as they are not specific for including affective expressions in characters.

Other tools have been proposed to allow students to express affect, like the Subtle Stone [2], Emotion Faces [21] and StoryFaces [33]. These works offer important examples of how to explore the communication of affect among students. The application we designed with children is also related, among other things, with the expression of affect in a learning context. We are interested in understanding what elements are likely to evoke more positive affective responses from students – that is, the elements that contribute to Affectability. Our goal is to suggest practical information on how to consider affect in the design process (and consequently in design results) of learning technology. Such technology can be any kind of application for learning environments and not only applications for dealing specifically with affect. The concept of Affectability was proposed and discussed in [15]. In the work reported here we further explore the concept by proposing the Design Principles and the Framework for Affectability, focusing on their practical application.

DESIGN FOR AFFECTABILITY

The Framework for Affectability is a conceptual framework for the design of digital technology for learning environments that helps designers to think about affect in the design process as well as in the product of this process. The framework adopts the structure of the Semiotic Onion [36], organizing the design process in three embedded layers: informal (the subculture where affect is socially constructed and experienced), formal (the methodologies and rationale of dealing with affect in design) and technical (the digital artifact, as a software application or hardware). Moreover, the framework adopts the socio-technical view of the Science of Design [3], as it shows the path of the design process starting in the society and going through the informal and formal layers of the Semiotic Onion, towards the creation of the digital technology and impacting back on society. The Principles for the Design for Affectability has basis on this design flow, which is organized in three columns: pre design, design and evaluation, as shown in Figure 1.

Design principles

The design Principles for Affectability aims at improved affective responses from children, as a result of their interaction with technology in learning environments. We derived the design principles both from *a posteriori* (empirical; derived from Participant Observation at a school) and *a priori* (theoretical; derived from literature review of preeminent authorities in psychology and education, focusing in their understanding of the role of affect in children's education) research. The results from one approach, confirmed the other and vice-versa, resulting in the design Principles for Affectability (PAf). We briefly name these principles as follows:

- *PAf.1 Support expression and interpretation of affective*

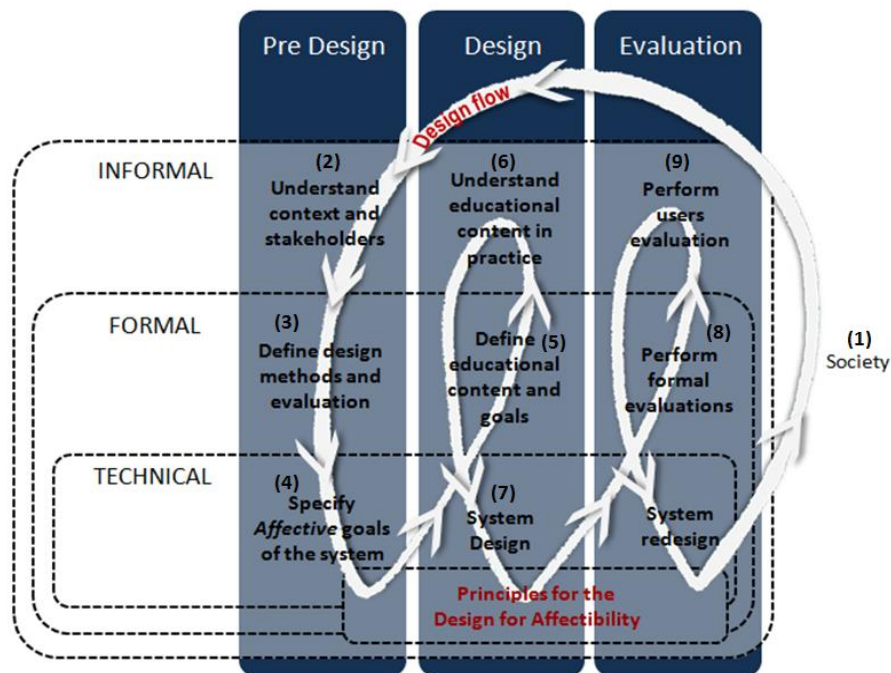


Figure 1 - Framework of the Design for Affectability.

states. The application should allow users to freely express and interpret affect.

- *PAf.2 Consider users' social context, including their values and culture.* Designers should take into account users' culture and values and make them clear in the designed application.
- *PAf.3 Allow tailoring.* Users should be able to tailor the application (e.g., with their own personal media or educational content) according to their needs or preferences.
- *PAf.4 Allow collaborative construction.* The application should allow users to collaborate with each others' work. Also some degree of adults' participation (e.g. of teachers and parents) could contribute to this construction.
- *PAf.5 Promote awareness.* The application should enable social awareness, letting users know the presence or status of peers.
- *PAf.6 Enable contamination by media.* The use of multimedia should be explicitly explored by designers in order to evoke certain moods during interaction.

These principles were the result of a work that had two fronts: theoretical review of the field of children psychology and education from affective perspective; and the observation (coded and categorized under qualitative techniques) from a two-year immersion at an elementary school in Brazil. Because of this approach, the principles can be recognized in schools' activities (with or without digital technology) that evoke rather positive affective responses from children. These principles find support in

the literature, as well, as briefly exemplified:

- *PAf.1:* A complete understanding of human thoughts can only be achieved when there is an understanding of the affective-volitional base (Vygotsky's view according to [40]).
- *PAf.2:* Values attributed to people are the starting point to feelings (Piaget's view according to [41]).
- *PAf.3:* "(...) Why not, for example, take advantage of the students' experience of life in (...) to discuss the problems of (...)" from Paulo Freire [42].
- *PAf.4:* Zone of Proximal Development, from Vygotsky [43].
- *PAf.5:* Affective exchanges are entirely dependent on the concrete presence of other partners (Wallon's view according to [44]).
- *PAf.6:* The highly contagious nature of emotion comes from the fact that it is visible, that it is open to the exterior by the modifications of mimics and facial expression (Wallon's view according to [44]).

Further information on the rationale of the Design Principles for Affectability, as well as practical examples of the principles in the design of technology can be found in [16].

THE FRAMEWORK INSTANTIATED

The approach that we propose with the Framework and Principles for Affectability has its feasibility illustrated in the process of development of an application to support learning in the context of an elementary public school in

Brazil. Next we briefly explain the instantiation of the Framework in our process. The numbers in parentheses refer to the steps of the design flow shown in Figure 1.

The design process started with the social aim – (1) society – of promoting some kind of benefit to the school community. During the pre-design phase, in order to (2) understand context and stakeholders, we were engaged in the daily activities of the school. This immersion provided us rich material that helped us to better understand the school's culture. This endeavor informed the next steps in the flow of design: we (3) defined design methods and evaluation, and (4) specified the affective goal of the application as well as its (5) educational goal. From the field observation, we noted that the students are prone to rather negative affective expressions at school. In an attempt to (4) promote rather positive expressions we suggested an application that would work similar to the Secret Santa activity ('amigo secreto' or 'amigo oculto' in Brazil), in which the gifts to be delivered would be compliments on the other person. The application would also (5) promote the practice – or perhaps even augmentation – of the vocabulary related to compliments (e.g., adjectives) and their spelling. To design such application we (3) defined as methods: participatory practices, complemented with the continued ethnographic approach that had already been in place at the school. The step (6) understanding educational content in practice also benefited from such ethnographic approach. The (7) system design was conducted with active participation of users during the Design Workshops that we present here, and other activities. In this iterative flow, the steps of (8) formal and (9) users evaluation are also represented in the Design Workshops as the evaluation of our preliminary prototype.

Design Workshops

We are calling Design Workshops the participatory practices, i.e., meetings that took place at the school with the active participation of the students, as part of the process of designing the proposed application. This paper focuses on steps from (6) until (9), informing the design and preliminary evaluation of an application. Next we describe the Design Workshops with its scenario, participants and procedures, as well as its results. Such description exemplifies the practical application of our framework, especially in relation to steps (3) and steps from (6) until (9).

Scenario and participants

The activities took place at an elementary public school located in a suburban area of a city in the state of São Paulo, Brazil. This school is located at a neighborhood considered of high vulnerability [37], i.e., exposed to risks related to poverty.

The school community comprehends around 530 people, including students, teachers and school staff. Students' age range from 6 to 16 years old and they are distributed in grades from the 1st until the 9th. These grades form the

elementary education in the Brazilian system. The school community has been making sense of the idea of having one laptop per child since 2009. More than 500 XO laptops from OLPC [30] were introduced at the school in 2010 and since then the school community has been incorporating the use of the laptops in their daily activities as an initiative from the XO Project [4].

All participants were aware of their participation in the research. Students' parents signed written consents. All documents and procedures were approved by the Committee of Ethics of our University. As other authors [14] investigated, we were also concerned with the impacts on children that results from their participation in the design process. According to Druin's description of the roles of children in design processes [7], our complete approach following the Framework for Affectability allowed not only the students, but also the entire school community, to grow from 'users', to 'testers', to 'informants' until actual 'design partners' in the processes.

While steps (1) and (2) from Figure 1 involved the entire school community in the major project [4], the steps that followed were conducted in smaller groups. Conducting the activities with all the 500 students would not be feasible due to time constraints. Because the application should be of interest of both younger and older children, we elected the 2nd and 8th graders as appropriate design partners. In total, around 60 students were involved, among the two groups of 2nd graders and one group of 8th graders. The fact that the age differences between those groups are big is exactly one of the reasons why we chose to work with them: we wanted to explore the variety of the contribution from diverse participants.

The workshops' rationale

Three Design Workshops were planned, with different focuses, as follows:

Breaking the Ice

Even though the students were already familiar with the presence of the researcher in their classrooms (due to the ethnographic approach, some students already referred to the researcher as their 'teacher'), an activity to 'break the ice' was still needed. The reason was that now they would have more active roles, participating as designers. Furthermore, besides the main researcher, other researchers would be present to register the activity in video and image and to provide help for the conduction of the Workshops.

Patterns of behavior

The objective of this Design Workshop was to identify the rules students use when conducting an activity similar to 'Secret Santa' without the use of technology. We were also interested in understanding the types of affect communication and expressions that take place during the activity. Understanding how an activity similar to 'Secret Santa' takes place without technology could inform the design of the digital application.



Figure 2 - Moments of the 'Breaking the ice' activity: The Little Theater of Speak [(a,b and c); and the Speak [33] running at a XO laptop (d).

Application Prototype

With student's results from the two previous Design Workshops, a preliminary non-functional prototype was assembled. We wanted to confirm previous results by checking whether the prototype represented what they imagined that the 'Secret Santa'-like activity would be in a digital format.

The main purpose of these meetings was to enable students to contribute as design partners in activities that should be suitable for them to freely express themselves. These three workshops were planned to also transit throughout the informal (breaking the ice), formal (patterns of behavior), and technical (application prototype) layers of the Semiotic Onion. Moreover, we were watchful for the role and the manifestation of the Principles for Affectability during the meetings and how the principles could be represented in the resulting application.

RESULTS

In this section we describe the activities and the results of the Design Workshops.

Breaking the ice

This Design Workshop lasted 1 hour and a half and had two parts. Each part was preceded by a warm up phase. In the first part, children would draw what they remembered from *Speak* (Figure 2d), an application from the XO Laptop [33]. In this application, the laptop's speech synthesizer "speaks" anything the user writes. It is presented in a simple interface depicting a face, whose eyes follow the mouse pointer and the mouth moves as it speaks. The user can control the

speed of the voice, as well as its accent. To prepare children for this drawing activity, as a warm up activity, first the researcher orally elicited from the children all the characteristics they remembered from *Speak*. After that, the children acted at what we called the "Little Theater of *Speak*". In it, they worked in pairs: one being the XO laptop performing the role of the *Speak*, and the other being the user itself. We prepared a cardboard with the image of the XO laptop screen as frame, where the children could insert their faces, as if wearing a mask, in order to play the role of the XO laptop face of *Speak* (see Figure 2). The user had to write a word on the XO and the other child would read it aloud. The word had to be a compliment. In preparation for this, the researcher elicited with the children words of compliment and wrote them on the blackboard. We conducted the same workshop twice, one for each group of 2nd graders.

As a result, we gathered: drawings, video with the role playing activities, and a list of the compliments that the children could remember.

Besides making the children familiar with the context of a Design Workshop, the activity provided us with some indications about how children understand and perceive the XO laptop and its systems, specially *Speak*. Children tried to act as robots, making an effort to suppress laughter or smiles and producing metallic voices (PAf.6). They tried to mimic the 'face' of the *Speak*, emphasizing the expressions and movement of their eyes.

Patterns of behavior



Figure 3 - One of the gifts (artistic expression of a compliment) created by the children and moments of the Design Workshop.

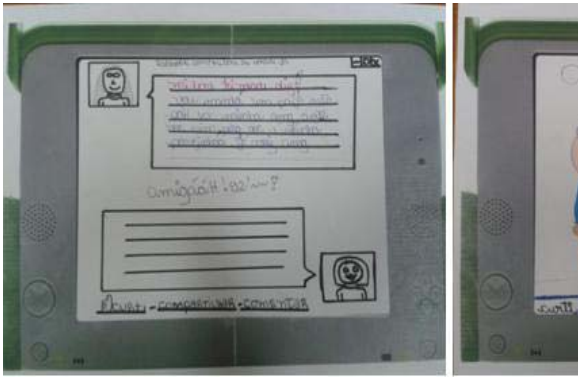


Figure 4 - Some of the results from the Design Workshop conducted with the 8th graders.

Since the intended application was the digital and modified version of 'Secret Santa', we prepared this Design Workshop with the objective of understanding details on how 'Secret Santa' happens without technology. We conducted the same workshop three times separately: with the two groups of 2nd graders and with one group of 8th graders.

In the first moment, we discussed with the entire groups about 'Secret Santa', asking whether they were familiar with the activity, how they played it, the problems that usually occur, how they solved them, and the variations that they knew of the same activity. After that, a real 'Secret Santa' took place. We raffled the names in the traditional way, with the names on pieces of paper and we asked the children to exchange compliments as gifts. We asked the children to imagine that the delivery of the gifts would take place within an application running at their laptops. In order to help them with this abstraction, we provided them with papers that had as a frame a colorful print of the XO laptop screen (see Figure 3b). Inside this frame, they should represent their compliments (i.e., their gifts) as well as the features that the application should have in order to support the delivery and the creation of the gift. Before they started creating their gifts, the researcher elicited with them some words for compliment. For 2nd graders this was their second time eliciting words.

Most of the children from the 2nd grade created drawings (PAf. 1 and 6) representing a person (either the person who would receive the gift, themselves, or both). Only one child from the 2nd grade added something more than the drawing: she wrote what she called a poem on it (PAf. 1). None of the 2nd graders represented interface elements. Drawings were also the dominant result among 8th graders. They, however, added interface elements (Figure 4). Among those elements there were buttons and icons for 'share', 'comment', 'chat', and 'like', mostly represented in a very similar way to the representation found in *Facebook* [8], commonly used among them (PAf. 2, 3 and 4). Three children wrote a song (PAf. 6). One of the children commented that she missed the exchange of secret notes among the participants (PAf. 1, 4 and 5), common along the

'Secret Santa' real activity.

This Design Workshop not only helped us to confirm the procedures for a 'Secret Santa', but also informed us about interface elements and the way children represent some terms. Figure 3 shows one of the resulting drawings (b) and moments of the activity (a and c).

The prototype

The objective of this Design Workshop was to confirm the ideas that were generated from children's works during the previous Design Workshops. This workshop took place three times: once with each of the two groups of 2nd graders and once with the group of 8th graders. Both groups were asked to look at the material they received and discuss about it using the online Chat application on their XO laptops. Due to the age differences, the activity varied slightly for each group.

We chose some drawings that resulted from the 'Secret Santa' Workshop and we gave these drawings for the children from the 2nd grade to talk about the interaction elements (e.g. icons, buttons, representations, etc.) that we had highlighted in the drawings. They discussed with each other, using the Chat application, about the meanings of those elements. The 8th graders interacted with a low fidelity prototype running on their XO laptops. This non-functional prototype was made using *Scratch* [24], following the Framework and Principles for Design for Affectability, as well as the input from our main design partners: the children.

We conducted informal interviews with the 2nd graders while they were participating in this Workshop. For the 8th graders, we gave them handouts with the print screens of the prototype so they could leave written comments on them. For both groups, interpersonal communicative skills were demanded, as they had to exchange ideas through written online interaction with peers. The log from the Chat from the 2nd graders was collected and transcribed, as well as the informal interviews with them. The researcher and the teacher decided to interrupt the online chat among the 8th graders. They were engaged in inappropriate subjects, generating uproar and loss of focus in the activity. Their written comments, as well as their spontaneous testimonials recorded in the video were used instead of the chat log. They reported enjoying the activity, especially the fact of being able to express their opinions. Most of the 8th graders did not understand that it was a low-fidelity prototype and complained that they were not able to actually use the features of the application (e.g. online chat, name raffling, secret message exchanging, etc.). Some of the features were available to interact, like the screen to compose the avatar. They had most fun creating the representation of their opinions (PAf. 1, 5 and 6). The 2nd graders, probably because of the format of the prototype (paper prototype), did not have problems understanding that it was not a real application. Even though they had not represented interface elements in their own drawings, they were able to

understand the meanings of the interface elements in the paper prototypes.

Design principles instantiated

The application that was being designed in these workshops should help students practice the vocabulary for words of compliments and the composition of other forms of expression like poems, songs or art. As a digital version of the ‘Secret Santa’ activity, the application shall manage the participants (students present at the school on a given day) and the interaction among them in online conversations with the entire group as well as exchange of individual messages without the disclosure of who the sender is. The application will let students compose the gift to be exchanged, allowing users to choose to use different media. The raffle without ties shall be controlled by the application. Moreover, students will be able to rate how they felt participating in the activity by composing avatars with facial expression and other features (e.g., shapes, accessories). These expressions should be convertible to grades and the grades from all participants could be stored and used by the system to compose graphs to indicate the history of the affective state of the students.

Figure 5 illustrates the application prototype that the 8th graders interacted and evaluated. It includes the results from the first two Design Workshops, as well as some of the Design Principles:

- Students can express and interpret affective states (PAf.1) via the online chat and secret message exchanging features. Moreover, they can create avatars that also express affective states.
- Students’ social context was accounted during the design process (PAf.2), as it strived to reflect students’ way of playing the ‘Secret Santa’ (‘amigo secreto’ or ‘amigo oculto’) and it used elements that were produced by students during the Design Workshops.
- Tailoring is allowed (PAf.3) as students can create their own avatars and the gifts (compliments) to be exchanged.
- The online chat and secret message exchanging features allow students to work together (PAf.4). Ideas can be exchanged and collective moods can be created.
- The application shows who are the students that are participating in the ongoing raffle of the ‘Secret Santa’ activity, allowing social awareness (PAf.5). Moreover, the avatars created should indicate the mood of each student as well as the mood of the entire group.
- Gifts can be created using different media formats (PAf.6). Students should be able to record songs or speeches, compose drawings, make short videos and take pictures in order to express their compliments to their peers.

DISCUSSION

In some fields, the main concern of interaction designers is to make applications that are faster and that generate fewer

mistakes. In such cases, the focus is on the final product only. When the context is related to children development, the focus is no longer on a single product but on the processes that take place in learning environments which the product supports. Other than being fast or accurate, applications have different desired characteristics that makes the process interesting, fun, exciting, supplies sensations of accomplishment and confidence – to name a few. This scenario demands new ways to think about design. The Framework and Principles for Design for Affectability can be one of the tools to support in this challenge.

Discussing at the informal level

Affect is a social and cultural product and the specificity of each social group demands methods that consider their uniqueness and rich backgrounds. Our proposed framework, based on the Semiotic Onion [36] and Science of Design [3], takes into consideration the actual context of children’s daily activities as the starting point of designing meaningful technology. From the informal level of children’s social context, the technology is developed and its use goes back into society, suggesting new practices, which affects the initial environment. We have noticed that not only the final designed application can have an impact in the social group; even before the deployment of the intended application, the Design Workshops have shown to be already an important opportunity for students to spend some time paying attention to positive characteristics of their peers. Our experience being among them, facing their reality, made us understand how even the apparently smaller and simpler actions can have an impact for marginalized communities. Merely our presence there already made a difference for those children, who were not accustomed to be the center of attention and concern. Being cared for is something they were not used to.

This awareness on the effect of the process as well as the final product on children puts in hands of designers a huge responsibility. Not only should the final designed applications be appropriate for students’ context, but also the process of design in which children are partners should be carefully planned. Children’s affective responses to the design process must also be of designers’ concern.

Discussing at the formal level

We mentioned that our Framework can be one of the tools to handle the challenge of designing for more positive affective responses. There are other interaction design tools and our proposal does not deny the importance of them, on the contrary. For example, usability and accessibility are extremely important concepts that can also impact users’ responses during their interaction with the application. The design for usability and accessibility should be default attributes to what we propose.

The approach we present in this paper brings to discussion the problem of Designing for Affectability and the importance of explicitly including the concern for affective results in the process of design. The report on the activities

and results from the Design Workshops could be taken as practical examples of a methodology that is aware of the vast and rich content from the informal level to bring them to the technology design, in a process performed together with the target audience.

Techniques like annotating, transcribing, coding, informal interviews and others were also used in our process. The details are not in the scope of this paper. However, the definition of the methods to be used in the design process is marked in step (3) of the Framework (Figure 1).

Discussing at the technical level

In previous section we have pointed out how the principles were used in our prototype. The principles, however, are highlighted inside the framework in a place that goes beyond the technical level. While they are reflected with

most concrete evidences in the final designed technology, the Design Principles for Affectability should be kept in mind throughout the entire process and layers.

Alternatively, designers might find useful the application of the Principles in a way detached from the Framework. One example would be the evaluation of existing technology and its redesign towards Affectability.

Discussing at an 'ecosystemic' level

There are some points that deserve special considerations: affect is a subjective concept; each social group has its particular culture and values; designers, be them the researchers or the children or the school community, are all actors in a continuously changing society.

The participation of the children in the design process

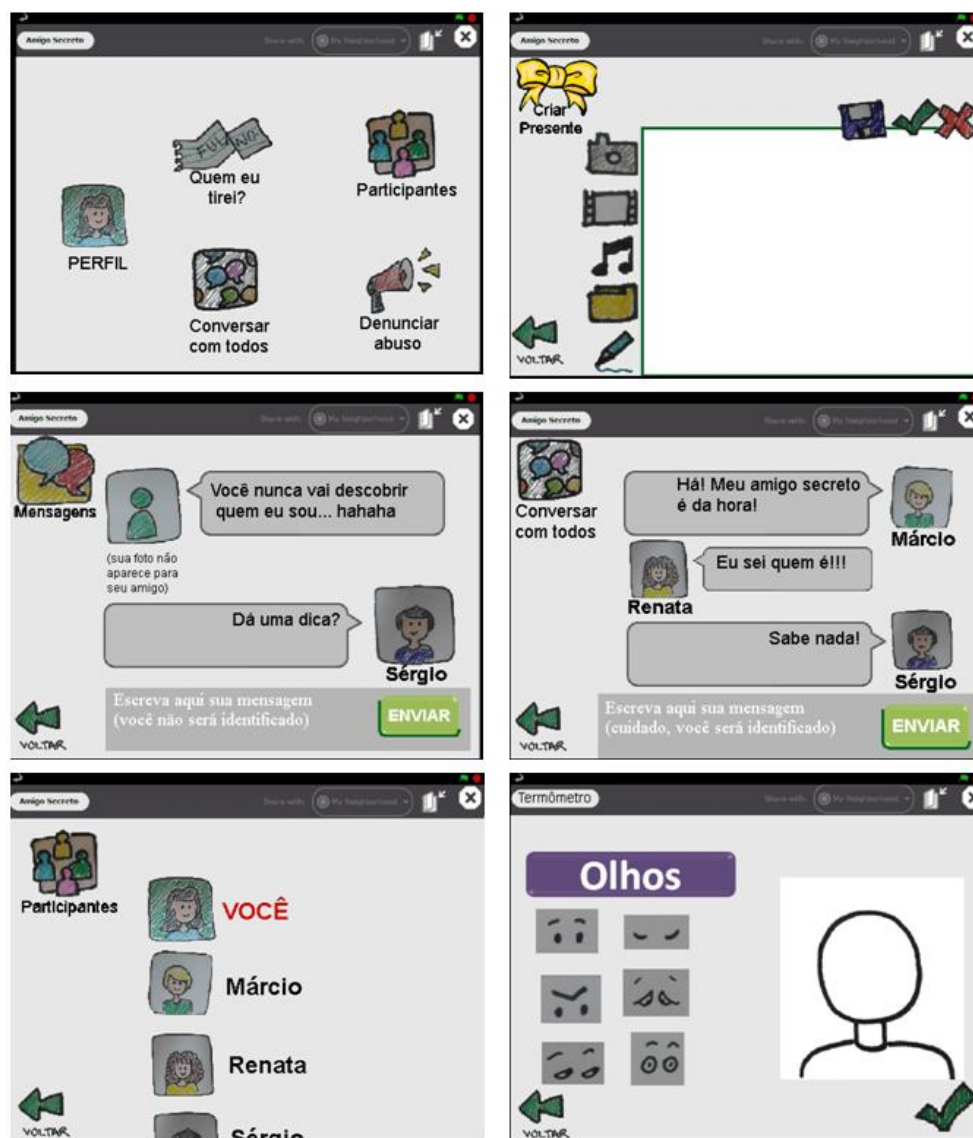


Figure 5 - Some screens of the low fidelity prototype used by the 8th graders during the Design Workshop. From left to right and top to bottom: initial screen; composition of the expression of compliment as a gift; individual secret messages; group chat room; list of participants; creation of an avatar.

might already contribute to the promotion of benefit that designers aim at the step (1) (from Figure 1), or change other parameters not noticed before. Moreover, each child has her/his own life, where other actors play important roles. For example, suppose a fight with her/his mother on the day of the activity, or a surprise party that happened right before the child went to school. These situations might influence how she/he perceives an application and therefore the assessment of affective response can be impacted.

One cannot guarantee that a set of recommendations will result in determined affective responses, of course [15]. However, as designers, we can create possibilities for uses to make it more likely for rather positive experiences to be evoked during the interaction. We can take actions towards more meaningful and sensible design.

CONCLUSION

Affect and emotion have been acknowledged as determinant aspects in children's development. There is still a lack of understanding on how to include affect in the practice of design and its resulting products. In this paper we approached the challenge of creating technology in a process in which affect is explicitly considered. This complex task demands new methods to support designers when projecting applications for (with and by) children. We presented our experience in conducting Design Workshops with children of the elementary years of the Brazilian educational system, in order to collaboratively build and evaluate the prototype of an application that allows them to express and interpret affect. The design process for creating this application follows the Framework and Principles for the Design for Affectability presented here. In order to better understand the impact of the design for Affectability in the resulting application, further work is going on involving the development of the application and the assessment of its impact on children in their school context. Moreover, the framework's validation with groups of designers is now in course.

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REFERENCES

1. Antle, A.N. (2007) The CTI Framework: Informing the Design of Tangible Systems for Children. In: Proc. of TEI'07, pp. 195-202.
2. Balaam, M.; Fitzpatrick, G.; Good, J.; Luckin, R. (2010) Exploring Affective Technologies for the Classroom with the Subtle Stone. In: Proc. of CHI'10, pp. 1623-1632.
3. Baranauskas, M. C. C. (2009) Socially Aware Computing. In: Proc. of ICECE 2009, pp. 1-5.
4. Baranauskas, M. C. C.; Martins, M. C.; Assis, R (2012) XO na Escola: construção compartilhada de conhecimento – lições aprendidas, NIED/UNICAMP.
5. Boehner, K., DePaula, R., Dourish, P., Sengers, P. (2007) How Emotion is Made and Measured. *International Journal of Human-Computer Studies*, 65 (4), pp. 275-291.
6. DeVries, R.: Piaget's Social Theory. *The Constructivist*, 17(1) (2006).
7. Druin, A. (2002) The role of children in the design of new technology. *Behaviour and Information Technology*, 21 (1), pp. 1-25.
8. Duh, H.B.; Yee, S.L.C.Y.; Gu, Y.X.; Chen, V.H. (2010) A Narrative-Driven Design Approach for Casual Games with Children. In: Proc of Sandbox 2010, pp.19-24.
9. Facebook. <http://www.facebook.com> (last accessed: April, 2013).
10. Frauenberger, C.; Good, J.; Alcorn, A.; Pain, H. (2012) Supporting the Design Contributions of Children With Autism Spectrum Conditions. In: Proc. of IDC'12, pp. 134-143.
11. Furtado, E.; Furtado, V.; Vaconcelos, E. (2007) A conceptual framework for the design and evaluation of affective usability in educational geosimulation systems. In: Proc. of INTERACT 2007, Rio de Janeiro, 2007, pp. 497-510.
12. Giaccardi, E.; Paredes, P.; Diaz, P.; Alvarado, D. (2012) Embodied Narratives: A Performative Co-Design Technique. In: Proc. of DIS'12.
13. Girard, S.; Johnson, H. (2011) Designing affective animations with children as design partners using role-playing. In: Proc. of IHM'11, article 26, 8 pages.
14. Guha, M.L.; Druin, A.; Fails, J.A. (2010) Investigating the Impact of Design Processes on Children. In: Proc. of IDC 2010, pp. 198-201.
15. Hayashi, E. C. S.; Baranauskas, M. C. C. (2011) The Affectability Concept in Systems for Learning Contexts. *International Journal for e-Learning Security (IJeLS)*, 1 (1/2), 10-18.
16. Hayashi, E.C.S.; Baranauskas, M.C.C (2013) Design Principles for Affectability. Technical Report IC-13-17.
17. Harper, R., Rodden, T., Rogers, Y., and Sellen, A. (2008) Being Human: Human-Computer Interaction in the Year 2020. Microsoft Research, Cambridge, U.K.
18. Hassenzahl, M. (2004) Emotions can be quite ephemeral: We cannot design them. *Interactions*, 11 (5), 46-48.
19. Hoonhout, J. (2008) Let's Start to Create a Fun Product: Where is My Toolbox? In: User Experience Evaluation Methods in Product Development Workshop in CHI'08.
20. Hornof, A.J. (2009) Designing with Children with Severe Motor Impairments. In: Proc. of CHI'09, pp. 2177-2180.
21. Humphries, L.; McDonald, S. (2011) Emotion Faces:

- The Design and Evaluation of a Game for Preschool Children. In: Proc. of CHI'11, pp. 1453-1458.
22. Isen, A., Daubman, K., and Nowicki, G. (1987) Positive Affect Facilitates Creative Problem Solving. *Journal of Personality and Social Psychology*, 52(6), 1122-1131.
 23. Kleinginna, P. R. ; Kleinginna, A. M. (1981) A categorized list of emotion definitions, with suggestions for a consensual definition. *Motivation and Emotion*, 5, p. 345-379.
 24. LaTaille, Y.(1992) Desenvolvimento do juízo moral e afetividade na teoria de Jean Piaget. Piaget, Vygotsky, Wallon: Teorias Psicogenéticas em Discussão. São Paulo: Summus Editorial.
 25. Lu, F.; Tian, F.; Jiang, Y.; Cao, X.; Luo, W; Li, G; Zhang, X.; Dai, G.; Wang, H. (2011) ShadowStory: creative and collaborative digital storytelling inspired by cultural heritage. In: Proc. of CHI'11, pp. 1919-1928.
 26. Maloney, J.; Resnick, M.; Rusk, N.; Silverman, B.; Eastmond, E. (2010) The Scratch Programming Language and Environment. In: *ACM Transactions on Computing Education*, 10 (4).
 27. Mazzone, E.; Iivari, N.; Tikkanen, R.; Read, J.C.; Beale, R. (2010) Considering Context, Content, Management, and Engagement in Design Activities with Children. In: Proc. of IDC'10, pp. 108-117.
 28. Millen, L.; Cobb, S.; Patel, H. (2011) A Method for Involving Children with Autism in Design. In: Proc. of IDC'11, pp. 185-188.
 29. Montemayor, J.; Druin, A.; Chipman, G.; Farber, A.; Guha, M.L. (2004) Tools for Children to Create Physical Interactive StoryRooms. In: *Computers in Entertainment*, 2 (1), pp. 1-24.
 30. OLPC. <http://one.laptop.org/> (last accessed: April, 2013)
 31. Ortony, A., Norman, D.A. & Revelle, W. (2005) Affect and Proto-affect in effective functioning. *Who Needs Emotions? The Brain Meets the Machine*. New York: Oxford University Press, pp. 173-202.
 32. Pandey, S.; Srivastava, S. (2011) Tiblo: A tangible learning aid for children with dyslexia. In: Proc. of DESIRE'11, pp. 211-220.
 33. Ryokai, K.; Raffle, H.; Kowalski, R. (2012) StoryFaces: Pretend-Play with Ebooks to Support Social-Emotional Storytelling. In: Proc. of IDC'12, pp. 125-133.
 34. Rosenkranz, M.A., Jackson, D.C., Dalton, K.M., Dolski, I., Ryff, C.D., Singer, B.H., Muller, D., Kalin, N.H., Davidson, R.J.: Affective style and in vivo immune response: neurobehavioral mechanisms. *Proc. Natl Acad Sci USA* 100:11148–11152 (2003).
 35. Speak. <http://wiki.laptop.org/go/Speak> (last accessed: April, 2013)
 36. Stamper, R.; Liu, K.; Hafkamp, M.; Ades, Y. (2000) Understanding the Roles of Signs and Norms in Organisations - A Semiotic Approach to Information System Design. In: *Journal of Behavior & Information Technology*, 19 (1), 15-27.
 37. Stolco, S., Almeida, L. C. Escolas municipais de Campinas e vulnerabilidade sociodemográfica: primeiras aproximações. *Revista Brasileira de Educação*, 16 (48), p. 663-814, 2011.
 38. UNESCO. ICT in Education. <http://www.unesco.org/new/en/unesco/themes/icts/> (last accessed: April, 2013)
 39. Xavier, R. A. C.; Garcia, F. E.; Neris, V. P. A. (2012) Decisões de design de interfaces ruins e o impacto delas na interação: um estudo preliminar considerando o estado emocional de idosos. In: Proc. of IHC'12, pp. 127-136.
 40. Oliveira, M. (1992) O problema da afetividade em Vygotsky. São Paulo: Summus Editorial.
 41. deSouza, M. T. C. C. (2011) As Relações entre Afetividade e Inteligência no Desenvolvimento Psicológico. *Psicologia: Teoria e Pesquisa*, 27(2), 249-254.
 42. Freire, P. (1998) *Pedagogy of Freedom: Ethics, Democracy and Civic Courage*. Rowman and Littlefield.
 43. Vygotski, L. (1978) *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press.
 44. Dantas, H. (1992) A afetividade e a construção do sujeito na psicogenética de Wallon. Piaget, Vygotsky, Wallon: Teorias Psicogenéticas em Discussão. São Paulo: Summus Editorial.